Exam 2

① Calculate the determinant of

$$\begin{bmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 7 & 6 & 3 \end{bmatrix}$$

② If A is a square matrix such that $A^3 = A$, what are the possible values of $\det A$?

③ Do vectors of the form $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x^2 + y^2 + z^2 \leq 1 \right\}$ form a subspace of $\mathbb{R}^3$? Please explain

④ Let A be a 5×6 matrix with rank $A = 4$
Find dim Row A, dim Col A, dim Nul A,
dim Row A^T, dim Col A^T, dim Nul A^T.

⑤ Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ -6 \end{bmatrix} \right\}$ be a basis for $\mathbb{R}^2$.
If $x = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$, find $[x]_{\mathcal{B}}$.

⑥ (Ex G) Let A be an $m \times n$ matrix and let $v_1, v_2, \ldots, v_p$ be vectors in $\mathbb{R}^n$. If $v_1, v_2, \ldots, v_p$ are linearly dependent, explain why the vectors $Av_1, Av_2, \ldots, Av_p$ must also be linearly dependent.